

Kai Gu

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Education

Bocconi University PhD in Marketing	Milan, Italy 2026 – Present
Bocconi University MSc in Data Science	Milan, Italy 2024 – 2026
Zhejiang University BEcon in Economics	Hangzhou, China 2020 – 2024

Research Interests

Substantive: Economics of AI, Economics of Digitization & Platforms, Media Economics

Methodological: Causal Inference, Computational Social Sciences, Multimodal Data

Working Papers

From Directory to Oracle: How AI Search Reshapes the Web's Attention Economy

Qiaoni Shi, Kai Zhu, and Kai Gu

Presented at: Bocconi Marketing Research Camp, May 2026; ISMS Marketing Science Conference, Lisbon, Portugal, June 2026

For two decades, the web's attention economy has rested on an implicit bargain: search engines help users find information and route traffic to the websites that produce it. AI search threatens to break this bargain by serving as an oracle that delivers synthesised answers rather than a directory that distributes attention. Using URL-level clickstream data on 45,386 US desktop panelists over October 2024–July 2025, we provide the first within-user, multi-shock empirical characterisation of how AI search reshapes web traffic. Four facts: (i) ChatGPT Search adoption was rapid and shock-driven, with three access expansions lifting search-enabled users $\sim 1.7\times$ to $\sim 3.3\times$; (ii) AI search is absorptive—only 5.4% of ChatGPT conversations produce any outbound click vs. 33.1% of Google queries, and 67% of ChatGPT users never click out at all (vs. 11% for Google), with a within-user referral-rate gap of ~ 30 percentage points; (iii) AI search referrals are systematically tilted toward reference, developer, and academic destinations and away from ad-supported and community content, and within a user-week ChatGPT reaches only ~ 2 unique domains vs. Google's ~ 14 ; (iv) exploiting two ChatGPT Search rollouts in a stacked event-study design, AI-search adoption causes a ~ 9 – 11% reduction in Google queries, concentrating in informational destinations (Wikipedia, GitHub, NYT) while sparing transactional ones (Amazon).

Governing AI Content Tools: The Efficiency–Diversity Trade-off in Platform Knowledge Ecosystems

Kai Zhu, Kai Gu, and Lorenzo Lupo

Presented at: 13th Wiki Workshop (Online), March 2026

Knowledge platforms increasingly deploy AI tools to accelerate content production, yet the ecosystem-level consequences of these deployment decisions remain poorly understood. We conduct a large-scale audit of AI-assisted content creation on Wikipedia, examining how machine translation (MT) relates to information volume and diversity in multilingual knowledge ecosystems. Using over half a million matched article pairs across ten language editions, we document an efficiency–diversity trade-off: relative to independently written articles, MT-assisted articles deliver substantially greater information volume and content completeness for target-language readers, yet contribute significantly less novel information to the cross-language knowledge ecosystem. MT-assisted articles exhibit lower semantic divergence from their English source articles and lower information breadth across language pairs and topic categories; the missing content is often culturally anchored—local context, regional perspectives, and cultural details that human editors naturally add. To map the trade-off frontier, we simulate ecosystem-level information volume and diversity outcomes across a range of MT adoption ratios, illustrating how aggregate efficiency and diversity co-evolve as platforms shift their deployment strategies. Our findings suggest the need for explicit governance frameworks that acknowledge trade-offs, monitor ecosystem-level outcomes, and design AI deployment strategies that preserve valued properties while capturing efficiency gains.

Conference Presentations

ISMS Marketing Science Conference, Lisbon, Portugal
Bocconi Marketing Research Camp
13th Wiki Workshop (Online)

June 2026
May 2026
March 2026

Skills

Languages: English, Mandarin

Programming: R, Python, L^AT_EX, Markdown